

Dollarization and Foreign Direct Investment in Latin America and the Caribbean

Chakrayuddh Kimsovanna

Regression and Multivariate Analysis Project

Dr. Patrick Brandt

University of Texas at Dallas

May 8, 2026

Introduction

This study examines the relationship between dollarization and foreign direct investment inflows within the context of Latin American and Caribbean economies, as many of these countries have maintained very close economic ties with the United States, where many have become very dollarized in the process. Dollarization, in this case, refers to the ratio of foreign currency deposits to the total deposits that each country has; in other words, it represents how much of the money held in the domestic banking system is held in foreign currency rather than the local currency (Baliño, 2003). As such, this research looks into whether countries with higher levels of dollarization receive higher FDI inflows as a percentage of their GDP. While the definition of dollarization does not specify which foreign currency is taken into account, it is reasonable to assume that much of the foreign-currency deposits in this region involve mostly U.S. dollars. Theoretically, dollarization can make financial transactions easier for foreign firms, as they may not wish to hold any local currency, especially in economically volatile countries in which their domestic currencies tend to fluctuate as a result of inflation. Conversely, a high level of dollarization can signal that there is indeed volatility in said country's economy, so foreign firms may hesitate to invest in it (Baliño, 2003). Regardless, the hypothesis of the study is that higher dollarization increases FDI inflows into these economies, although the overall impact may be ambiguous without concrete empirical evidence.

Methodology

Data Collection and Defining Variables

The primary data sources are the Inter-American Development Bank (IADB) and the World Bank. The IADB data are used to measure dollarization, as it already includes the percentage of the foreign-currency deposits in the total deposits. Countries that are listed in the IADB data will be considered for this research, which include Argentina, the Bahamas, Jamaica, Panama, Uruguay, Venezuela, Barbados, Paraguay, Peru, Suriname, Trinidad and Tobago, Belize, Costa Rica, the Dominican Republic, Ecuador, Bolivia, Brazil, Chile, Colombia, El Salvador, Mexico,

Nicaragua, Guatemala, Guyana, Haiti and Honduras. This dataset has an interval from 1990 to 2023. Additionally, the World Bank API (via the WDI package) was used to pull data on various macroeconomic indicators appropriate for this study—among which are the FDI inflows. The dependent variable is FDI inflows as a percentage of the country's GDP, as measuring in terms of ratio would make it easier to compare countries with different economy sizes. The main independent variable is dollarization, expressed as a rate of foreign-currency deposits over the total amount of deposits, which is readily provided by the IADB. Control variables were also considered, such as the exchange rate (i.e., local currency units per U.S. dollar), trade openness (i.e., trade volume as a percentage of GDP), as well as inflation and GDP growth—both of which are in percentages.

Upon examining the datasets, neither Barbados nor Trinidad and Tobago have trade data, thus these two countries were dropped from the study. Another issue is that Argentina has records on inflation only from 2019 onwards, and its dollarization value in 2018 is missing entirely from the IADB report; likewise, Nicaragua is missing this data in 2018 and 2019, too. In 2023, 15 of the 18 countries (i.e., Argentina, the Bahamas, Belize, Bolivia, Chile, Costa Rica, the Dominican Republic, Ecuador, El Salvador, Honduras, Mexico, Nicaragua, Panama, Paraguay, and Uruguay) do not have their dollarization rates reported, which necessitates dropping 2023 from the datasets entirely. In both the IADB and World Bank datasets, for all countries, with the exception of Argentina for inflation, all variables are available from at least 2004 onward. It should also be noted that the datasets are structured as country-year panel data, so a one-year lag was applied to all explanatory variables, because FDI inflows may vary based on the previous year's macroeconomic conditions, too. As a result, the main sample covers 16 countries, from 2005 to 2022, with 283 observations. Inflation is not included in the main model because Argentina only has inflation records for four years, which would remove much of the country's sample from the model, and since Argentina is one of the major economies in Latin America, dropping it like the two aforementioned island countries would not be ideal. However, inflation is included in a secondary model as a way to check for robustness, which contains 270 observations.

Selecting the Model

The main model can be written as:

$$FDI_{it} = \beta_0 + \beta_1 Dollarization_{i,t-1} + \beta_2 ExchangeRate_{i,t-1} + \beta_3 Trade_{i,t-1} + \beta_4 GDPGrowth_{i,t-1} + \lambda_t + u_i + \epsilon_{it}$$

where the indexes i and t represent countries and years. FDI_{it} refers to FDI inflows as a percentage of GDP for country i in year t , while $Dollarization_{i,t-1}$ is the one-year lagged dollarization rate, $ExchangeRate_{i,t-1}$ is the one-year lagged exchange rate, $Trade_{i,t-1}$ is the one-year lagged trade openness measure, and $GDPGrowth_{i,t-1}$ is the one-year lagged GDP growth rate. The terms λ_t and u_i are the year effects and the country-level random effects, respectively, as ϵ_{it} is the remaining error term. Due to the panel nature of the data, a pooled OLS model would not be appropriate because it does not account for potential differences among countries that may be omitted from the included variables; for example, some countries can differ from each other in terms of economic regulations that may drive away foreign capital, histories of inflation and economic crises, conflicts and political instability, labor market conditions, as well as geographic location and regional trade agreements—all of which may affect FDI inflows, but including every single one of these factors would be beyond the scope of this study. This is why fixed effects and random effects models were more applicable, as the former allows each country to essentially have its own intercept, while the latter turns these omitted differences into random components instead. Similarly, year effects were considered to factor in various shocks that affect these countries in the same year, such as the financial crisis of 2008 and the COVID-19 pandemic. The Hausman test is then used to compare whether the random effects specification is acceptable relative to the fixed effects one.

Table 1: Hausman Test Result for the Main Two-Way Panel Model

Model	Chi-square	df	p-value
Main model	0.5001	4	0.9735

The p-value is above 0.05, based on this Hausman test, thus the random effects specification is not

rejected and the chosen model utilizes two-way random effects.

Performing Robustness Checks

Before making any inference from the model, it is essential to perform robustness checks and diagnostic tests to ensure the validity of the results. A VIF test was first conducted to check for multicollinearity among the variables. As shown in Table 2, all VIF values for each of the four explanatory variables of the main model are very close to 1, which means that there is no concern for multicollinearity.

Table 2: VIF Results for the Main Model

Variable	VIF
Dollarization	1.0375
Exchange rate	1.0005
Trade openness	1.0851
GDP growth	1.0596

Some more tests were performed to check for heteroskedasticity, serial correlation, and cross-sectional dependence. As reported in Table 3, the Breusch-Pagan test reveals a p-value of 0.0154, which is statistically significant and shows that there is a heteroskedasticity problem in the model. This issue can be counteracted by using clustered robust standard errors when making inferences, because they allow for error variance to differ across observations, and are also more appropriate for panel data when clustered by country. The panel serial correlation test also has a statistically significant p-value of less than 0.01, which further indicates there are issues with how the errors are correlated with each other, too, within countries over time; and one way to address this is to estimate the model with an AR(1) specification. The Pesaran CD test also shows evidence that there is cross-sectional dependence among the errors across countries; although this test does not warrant further modification to the model, it demonstrates that a two-way specification with time effects was necessary, even if it may not address the issue completely.

Table 3: Diagnostic Test Results for the Main Two-Way Random Effects Model

Diagnostic test	Statistic	df	p-value
Breusch-Pagan test	12.28	4	0.0154
Panel serial correlation test	107.62	15	4.611×10^{-16}
Pesaran CD test	5.0043	—	5.608×10^{-7}

After estimating the two-way random effects model with AR(1) specification, the normalized residuals were graphed on the ACF plot. Aside from the lag 0 where autocorrelation is always 1.0, most of the lags have autocorrelation values well within the blue-dotted confidence bands. There are a few exceptions at lags 6, 14 and 18, although these crossings are relatively minuscule (Figure 1). Essentially, the AR(1) specification does appear to account for most of the serial correlation in the model.

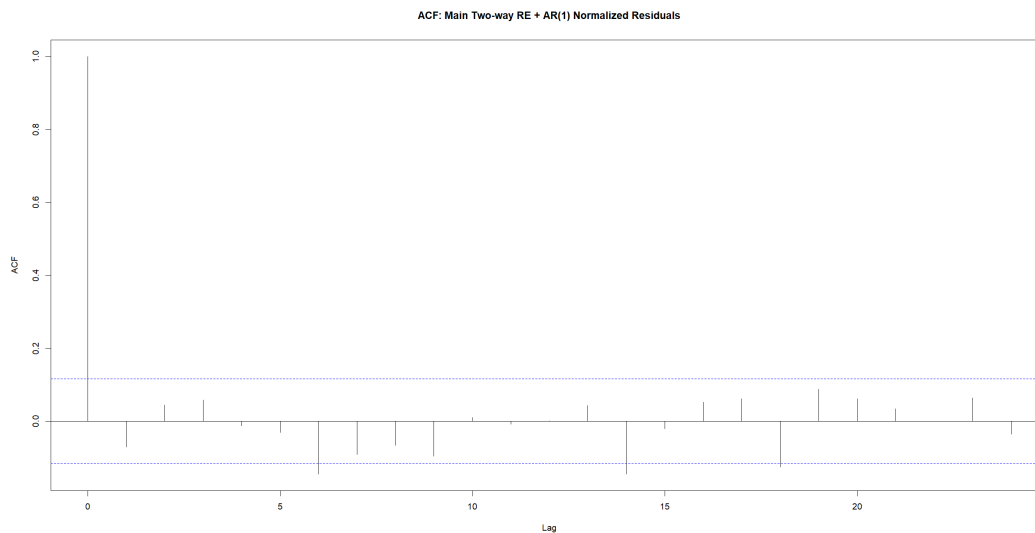


Figure 1: ACF Plot of Normalized Residuals for the Main Two-Way RE + AR(1) Model

More robustness check was done by adding lagged inflation as another control variable to see if the relationship between dollarization and FDI inflows still hold; the new secondary model still utilizes the same specifications as the main model, but the sample size is now reduced to 270 observations. This variable is represented as $Inflation_{i,t-1}$, as the equation becomes:

$$FDI_{it} = \beta_0 + \beta_1 Dollarization_{i,t-1} + \beta_2 ExchangeRate_{i,t-1} + \beta_3 Trade_{i,t-1} \\ + \beta_4 GDPGrowth_{i,t-1} + \beta_5 Inflation_{i,t-1} + \lambda_t + u_i + \epsilon_{it}.$$

Table 4: Diagnostic Results for the Inflation-Extended Model
VIF Results

Variable	VIF	<i>Diagnostic Tests</i>		
		Test	Statistic	p-value
Dollarization	1.0235			
Exchange rate	1.0025	Breusch-Pagan	9.9965	0.0753
Trade openness	1.0930	Serial correlation	92.5390	$< 2.2 \times 10^{-16}$
GDP growth	1.0809	Pesaran CD	3.5936	0.0003
Inflation	1.0147			

This inflation-extended model shares similar VIF results to the main model, so there is no evidence of multicollinearity. Meanwhile, the inflation-extended model has similar problems regarding serial correlation and cross-sectional dependence, as their respective tests show statistically significant p-values. Interestingly, the Breusch-Pagan test shows a p-value of 0.0753, meaning that there is no strong evidence that there is a heteroskedasticity issue with the secondary model. While this means that clustered robust standard errors are not required, they could still be used for consistency when comparing with the main model's results. The ACF plot also shows that aside from lags 6 and 14, the residual autocorrelations are all within the confidence bands, so the AR(1) specification also performs well with this new variable added (Figure 2).

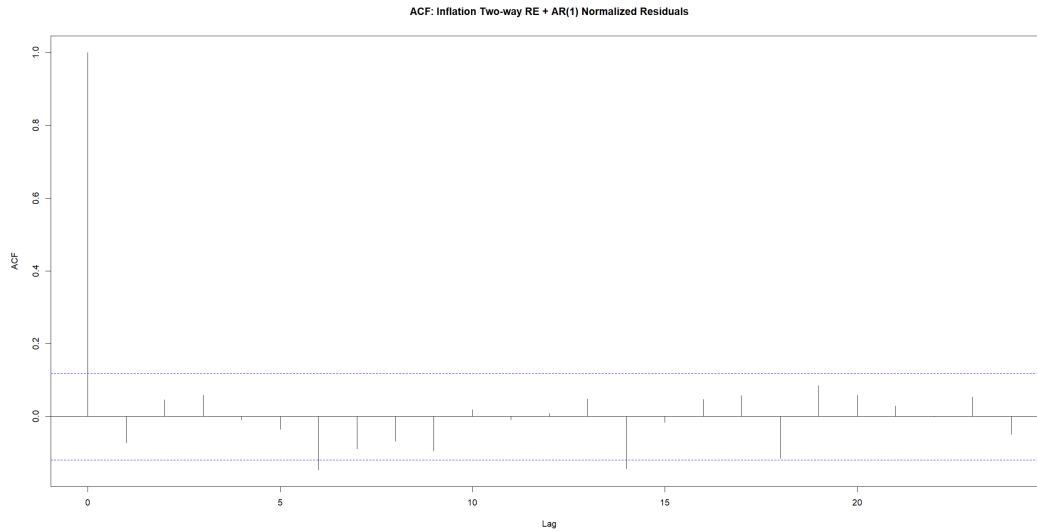


Figure 2: ACF Plot of Normalized Residuals for the Inflation-Extended Two-Way RE + AR(1) Model

Results

The final results are based on the two-way random effects AR(1) models that also use country-clustered robust standard errors. Both main and inflation-extended models produce similar coefficient patterns (Table 5). Most importantly, based on the two models, it is clear that dollarization does not have any statistically significant association with FDI inflows, as their p-values are 0.650 and 0.600, respectively. The coefficient estimates for dollarization are negative, but one cannot state that higher dollarization decreases FDI inflows because these estimates are not statistically significant. There is also no evidence that exchange rates and GDP growth across both models are associated with any change in FDI inflows, although they, too, have negative coefficients. Likewise, the added inflation variable in the second model does not have a statistically significant effect either.

Table 5: Two-Way RE+AR(1) Results with Country-Clustered Robust Standard Errors

Variable	Main Model	Inflation-Extended Model
Dollarization	-0.0046534 (0.009936) [0.64989]	-0.0055552 (0.010238) [0.59980]
Exchange rate	-0.0003659 (0.000395) [0.50424]	-0.0003691 (0.000397) [0.50308]
Trade openness	0.0543763** (0.011561) [0.00264]	0.0527761** (0.012241) [0.00437]
GDP growth	-0.0283742 (0.036276) [0.45475]	-0.0177889 (0.036427) [0.63900]
Inflation	– – –	-0.0110566 (0.021863) [0.65635]
Year effects	Yes	Yes
Random country effects	Yes	Yes
AR(1) error structure	Yes	Yes
Country-clustered robust SEs	Yes	Yes
Observations	283	270
Countries	16	16
AR(1) phi	0.3721581	0.3733666

Note. Country-clustered robust standard errors are reported in parentheses. P-values are reported in brackets. The dependent variable is FDI inflows as a percentage of GDP. All explanatory variables are lagged by one year. Year effects are included but omitted from the table. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

However, what remains consistent across both models is that trade openness (i.e., trade volume as a share of GDP) is statistically significant with a p-value of 0.003 in the main model and 0.004 in the inflation-extended model. In the main model, the coefficient estimate is about 0.0544, which means that if trade openness increases by 1 percentage point, FDI inflows are predicted to increase by about 0.0544 percentage points of GDP. When taking inflation into account, the coefficient estimate for trade openness changes to approximately 0.0528, meaning FDI inflows are estimated to increase by 0.0528 percentage points of GDP as trade openness rises by 1 percentage point.

Given the results, while dollarization alone is not empirically tied to an increase in FDI inflows, one cannot fully rule out its relevance, as it may be one component among various economic factors (such as bilateral trade agreements, economic infrastructure, labor costs, natural resource abundance, etc.) that, when combined together, would have a more tangible influence on the dynamics of FDI. As stated earlier, the effects of dollarization can go both ways in that it may ease the transaction processes for foreign firms and provide them with reassurance in case of economic volatility, but it may also trigger the perception that the country they want to invest in could be unstable and have a monetary system weak enough for it to rely heavily on outside economies, which would make investors less willing to invest. One possible reason why trade openness is significant is that FDI is already closely tied to globalization, as economies that trade more (with fewer barriers) are more connected to international markets, thereby allowing foreign companies to invest more easily in said economies.

Conclusion

Overall, the results do not support the hypothesis that higher dollarization leads to an increase in FDI inflows. Across both the main and inflation-extended models, dollarization is statistically insignificant, as it alone is not a reliable predictor of FDI inflows. On the other hand, trade openness is the only variable that is statistically significant and has a positive coefficient, shown in both models. Nevertheless, there is a possibility that dollarization itself can be endogenous, as countries may choose to become more dollarized due to preexisting conditions like high inflation

and a weak monetary system. Future research could address the issue here by possibly adding more control variables, utilizing instrumental variables, separating fully dollarized countries from partially dollarized countries, or using synthetic control methods for those that have any particular events where the U.S. dollar is officially adopted.

References

- Baliño, T. J. T. (2003). Chapter 30: Dollarization: A primer. In *Current developments in monetary and financial law* (Vol. 2). International Monetary Fund. <https://doi.org/10.5089/9781589061767.072.ch030>
- Inter-American Development Bank. (n.d.). *Latin Macro Watch dataset*. IDB Data. Retrieved May 7, 2026, from <https://data.iadb.org/dataset/latin-macro-watch-dataset/resource/d450ac41-997f-4453-86a2-3bdccba2f1c3>.
- World Bank. (n.d.). *World Development Indicators*. World Bank Open Data. Retrieved May 7, 2026, from <https://databank.worldbank.org/source/world-development-indicators>.

Appendix for R Codes

Data Collection and Processing

```

1  ## <<<<<<<< Setting Up >>>>>>>>
2  rm(list=ls()) # Clear environment
3  oldpar <- par() # save default graphical parameters
4  if (!is.null(dev.list()["RStudioGD"])) # Clear plot window
5      dev.off(dev.list()["RStudioGD"])
6  cat("\014")
7
8  packages <- c(
9      "tidyverse", # for data cleaning and ggplot
10     "janitor", # for cleaning variable names
11     "plm", # for panel models and FE/RE
12     "lmtest", # for tests and coeftest
13     "sandwich", # for robust/clustered SEs
14     "car", # for VIF
15     "modelsummary", # for making regression tables
16     "nlme", # for AR(1) error models
17     "clubSandwich" # for clustered SEs
18 )
19
20 installed <- rownames(installed.packages())
21 for (p in packages) {
22     if (!(p %in% installed)) {
23         install.packages(p)
24     }
25 }
26
27 lapply(packages, library, character.only = TRUE)
28
29 ## <<<<<<<< Getting Dollarization Data >>>>>>>>
30 data_dollarization <- read_csv("foreign_currency_deposits_percentage_of_total_depo.csv")
31
32 data_dollarization_annual <- data_dollarization %>%
33     filter(frequency == "Annual") %>%
34     transmute(
35         isoalpha3,
36         country,
37         year = as.integer(period),
38         dollarization = as.numeric(value)

```

```

39   )
40
41   data_dollarization_annual %>%
42     summarise(
43       n_obs = n(),
44       n_countries = n_distinct(isoalpha3),
45       min_year = min(year, na.rm = TRUE),
46       max_year = max(year, na.rm = TRUE)
47     )
48
49   ## <<<<<<<< Getting World Bank Data >>>>>>>>
50   countries <- unique(data_dollarization_annual$isoalpha3)
51
52   wb_indicators <- c(
53     fdi_gdp = "BX.KLT.DINV.WD.GD.ZS", # FDI net inflows (% of GDP)
54     exchange_rate = "PA.NUS.FCRF", # official exchange rate, LCU per USD
55     trade_gdp = "NE.TRD.GNFS.ZS", # trade (% of GDP)
56     inflation = "FP.CPI.TOTL.ZG", # inflation, consumer prices (annual %)
57     gdp_growth = "NY.GDP.MKTP.KD.ZG" # GDP growth (annual %)
58   )
59
60   data_wb <- WDI(
61     country = countries,
62     indicator = wb_indicators,
63     start = min(data_dollarization_annual$year, na.rm = TRUE),
64     end = max(data_dollarization_annual$year, na.rm = TRUE),
65   )
66
67   data_wb_clean <- data_wb %>%
68     rename(
69       isoalpha3 = iso3c,
70       country_wb = country
71     ) %>%
72     select(
73       isoalpha3,
74       country_wb,
75       year,
76       fdi_gdp,
77       exchange_rate,
78       trade_gdp,
79       inflation,

```

```

80     gdp_growth
81   )
82
83   ## <<<<<<<<< Merging the Two Datasets >>>>>>>>>
84   data_dollarization_wb <- data_dollarization_annual %>%
85     left_join(data_wb_clean, by = c("isoalpha3", "year"))
86   data_dollarization_wb <- data_dollarization_wb %>%
87     select(-country_wb)
88
89   data_dollarization_wb %>%
90     summarise(
91       n_obs = n(),
92       n_countries = n_distinct(isoalpha3),
93       min_year = min(year),
94       max_year = max(year),
95       missing_dollarization = sum(is.na(dollarization)),
96       missing_fdi = sum(is.na(fdi_gdp)),
97       missing_exchange = sum(is.na(exchange_rate)),
98       missing_trade = sum(is.na(trade_gdp)),
99       missing_inflation = sum(is.na(inflation)),
100      missing_growth = sum(is.na(gdp_growth))
101    ) %>%
102    print(width = Inf)
103
104   ## <<<<<<<<< Checking for Earliest Available Year for Each Variable by Country >>>>>>>>>
105   data_earliest_data <- data_dollarization_wb %>%
106     group_by(country, isoalpha3) %>%
107     summarise(
108       dollarization_start = min(year[!is.na(dollarization)], na.rm = TRUE),
109       fdi_start           = min(year[!is.na(fdi_gdp)], na.rm = TRUE),
110       exchange_start      = min(year[!is.na(exchange_rate)], na.rm = TRUE),
111       trade_start         = min(year[!is.na(trade_gdp)], na.rm = TRUE),
112       inflation_start      = min(year[!is.na(inflation)], na.rm = TRUE),
113       growth_start        = min(year[!is.na(gdp_growth)], na.rm = TRUE),
114       .groups = "drop"
115     )
116
117   data_earliest_data
118   ### Neither Barbados nor Trinidad and Tobago have any data on trade, so they will be dropped.
119   ### Argentina has inflation data only from 2019 onwards.
120   ### For all other countries, all variables are available from at least 2004 onward.

```

```

121
122 ## <<<<<<<< Dropping the Two Island-Countries >>>>>>>>>
123 data_dollarization_wb_main <- data_dollarization_wb %>%
124   filter(!isoalpha3 %in% c("BRB", "TTO")) %>%
125   filter(year >= 2004)
126
127 data_dollarization_wb_main %>%
128   summarise(
129     n_obs = n(),
130     n_countries = n_distinct(isoalpha3),
131     min_year = min(year),
132     max_year = max(year)
133   )
134
135 ## <<<<<<<< Checking for NA >>>>>>>>>
136 na_locations <- data_dollarization_wb_main %>%
137   pivot_longer(
138     cols = c(dollarization, fdi_gdp, exchange_rate, trade_gdp, inflation, gdp_growth),
139     names_to = "variable",
140     values_to = "value"
141   ) %>%
142   filter(is.na(value)) %>%
143   select(country, isoalpha3, year, variable) %>%
144   arrange(country, year, variable)
145
146 na_locations
147 ### The only missing values are the inflation records for Argentina from 2004 to 2017.
148
149
150 expected_panel <- expand_grid(
151   isoalpha3 = unique(data_dollarization_wb_main$isoalpha3),
152   year = 2004:2023
153 )
154
155 missing_country_years <- expected_panel %>%
156   anti_join(
157     data_dollarization_wb_main %>% select(isoalpha3, year),
158     by = c("isoalpha3", "year")
159   ) %>%
160   left_join(
161     data_dollarization_wb_main %>% distinct(isoalpha3, country),

```



```

162     by = "isoalpha3"
163   ) %>%
164   arrange(country, year)
165
166 missing_country_years
167
168 ### Fifteen countries have missing dollarization data in 2023, so 2023 would not be considered.
169
170 data_dollarization_wb_main <- data_dollarization_wb %>%
171   filter(!isoalpha3 %in% c("BRB", "TTO")) %>%
172   filter(year >= 2004, year <= 2022)
173
174 ## <<<<<<<< Creating Two Versions of Data >>>>>>>>
175 data_model_main <- data_dollarization_wb_main %>%
176   drop_na(fdi_gdp, dollarization, exchange_rate, trade_gdp, gdp_growth)
177 data_model_with_inflation <- data_dollarization_wb_main %>%
178   drop_na(fdi_gdp, dollarization, exchange_rate, trade_gdp, inflation, gdp_growth)
179
180 data_model_main %>%
181   summarise(
182     n_obs = n(),
183     n_countries = n_distinct(isoalpha3),
184     min_year = min(year),
185     max_year = max(year)
186   )
187
188 data_model_with_inflation %>%
189   summarise(
190     n_obs = n(),
191     n_countries = n_distinct(isoalpha3),
192     min_year = min(year),
193     max_year = max(year)
194   )
195
196 data_model_main %>%
197   group_by(year) %>%
198   summarise(
199     n_countries = n_distinct(isoalpha3),
200     countries = paste(sort(unique(isoalpha3)), collapse = ", "),
201     .groups = "drop"
202   ) %>%

```

```

203   arrange(year)
204
205 data_model_with_inflation %>%
206   group_by(year) %>%
207   summarise(
208     n_countries = n_distinct(isoalpha3),
209     countries = paste(sort(unique(isoalpha3)), collapse = ", "),
210     .groups = "drop"
211   ) %>%
212   arrange(year)
213
214 setdiff(2004:2022, unique(data_model_main$year))
215 setdiff(2004:2022, unique(data_model_with_inflation$year))
216
217 ## <<<<<<<< Saving the Files >>>>>>>>
218 write_csv(data_dollarization_wb_main, "data_dollarization_wb_main.csv")
219 write_csv(data_model_main, "data_model_main.csv")
220 write_csv(data_model_with_inflation, "data_model_with_inflation.csv")

```

Model Estimation and Diagnostics

```

1  ## <<<<<<<< Pulling Data and Applying Lags >>>>>>>>
2  data_model_main <- read_csv("data_model_main.csv") %>%
3    arrange(isoalpha3, year) %>%
4    group_by(isoalpha3) %>%
5    mutate(
6      dollarization_lag = dplyr::lag(dollarization, 1),
7      exchange_rate_lag = dplyr::lag(exchange_rate, 1),
8      trade_gdp_lag = dplyr::lag(trade_gdp, 1),
9      gdp_growth_lag = dplyr::lag(gdp_growth, 1),
10     year_lag = dplyr::lag(year, 1)
11   ) %>%
12   mutate(
13     dollarization = if_else(year - year_lag == 1, dollarization_lag, NA_real_),
14     exchange_rate = if_else(year - year_lag == 1, exchange_rate_lag, NA_real_),
15     trade_gdp = if_else(year - year_lag == 1, trade_gdp_lag, NA_real_),
16     gdp_growth = if_else(year - year_lag == 1, gdp_growth_lag, NA_real_)
17   ) %>%
18   ungroup() %>%
19   select(-dollarization_lag, -exchange_rate_lag, -trade_gdp_lag, -gdp_growth_lag, -year_lag) %>%
20   drop_na(fdi_gdp, dollarization, exchange_rate, trade_gdp, gdp_growth)

```

```

21
22 data_model_with_inflation <- read_csv("data_model_with_inflation.csv") %>%
23   arrange(isoalpha3, year) %>%
24   group_by(isoalpha3) %>%
25   mutate(
26     dollarization_lag = dplyr::lag(dollarization, 1),
27     exchange_rate_lag = dplyr::lag(exchange_rate, 1),
28     trade_gdp_lag = dplyr::lag(trade_gdp, 1),
29     gdp_growth_lag = dplyr::lag(gdp_growth, 1),
30     inflation_lag = dplyr::lag(inflation, 1),
31     year_lag = dplyr::lag(year, 1)
32   ) %>%
33   mutate(
34     dollarization = if_else(year - year_lag == 1, dollarization_lag, NA_real_),
35     exchange_rate = if_else(year - year_lag == 1, exchange_rate_lag, NA_real_),
36     trade_gdp = if_else(year - year_lag == 1, trade_gdp_lag, NA_real_),
37     gdp_growth = if_else(year - year_lag == 1, gdp_growth_lag, NA_real_),
38     inflation = if_else(year - year_lag == 1, inflation_lag, NA_real_)
39   ) %>%
40   ungroup() %>%
41   select(-dollarization_lag, -exchange_rate_lag, -trade_gdp_lag, -gdp_growth_lag, -inflation_lag,
42     -year_lag) %>%
43
44   drop_na(fdi_gdp, dollarization, exchange_rate, trade_gdp, gdp_growth, inflation)
45
46
47 data_model_main %>%
48   summarise(
49     observations = n(),
50     countries = n_distinct(isoalpha3),
51     min_year = min(year),
52     max_year = max(year)
53   )
54
55 data_model_with_inflation %>%
56   summarise(
57     observations = n(),
58     countries = n_distinct(isoalpha3),
59     min_year = min(year),
60     max_year = max(year)

```

```

61   )
62
63   ## <<<<<<<<< Creating Panel Data >>>>>>>>>
64   pdata_main <- pdata.frame(data_model_main, index = c("isoalpha3", "year"))
65   pdata_inflation <- pdata.frame(data_model_with_inflation, index = c("isoalpha3", "year"))
66
67   ## <<<<<<<<< Main Model >>>>>>>>>
68   main_variables <- fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth
69
70   model_main_ols <- plm(
71     main_variables,
72     data = pdata_main,
73     model = "pooling"
74   )
75
76   ## Oneway FE/RE Effects
77   model_main_fe_oneway <- plm(
78     main_variables,
79     data = pdata_main,
80     model = "within",
81     effect = "individual"
82   )
83
84   model_main_re_oneway <- plm(
85     main_variables,
86     data = pdata_main,
87     model = "random",
88     effect = "individual"
89   )
90
91   summary(model_main_ols)
92   summary(model_main_fe_oneway)
93   summary(model_main_re_oneway)
94
95   hausman_main_oneway <- phtest(model_main_fe_oneway, model_main_re_oneway)
96   hausman_main_oneway
97   ### p-value is 0.9783, so RE can be used here for main model (one-way)
98
99   ## Twoway FE/RE Effects
100  model_main_fe_twoway <- plm(
101    main_variables,

```

```

102   data = pdata_main,
103   model = "within",
104   effect = "twoways"
105 )
106
107 model_main_re_twoway <- plm(
108   main_variables,
109   data = pdata_main,
110   model = "random",
111   effect = "twoways"
112 )
113
114 summary(model_main_fe_twoway)
115 summary(model_main_re_twoway)
116
117 hausman_main_twoway <- phtest(model_main_fe_twoway, model_main_re_twoway)
118 hausman_main_twoway
119 ### p-value is 0.9735, so RE can be used here for main model (two-way)
120
121 ##=====
122
123 ## <<<<<<<< Inflation Model >>>>>>>>
124 inflation_variables <- fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth +
125   inflation
126
127 model_inflation_ols <- plm(
128   inflation_variables,
129   data = pdata_inflation,
130   model = "pooling"
131 )
132
133 ## Oneway FE/RE Effects
134 model_inflation_fe_oneway <- plm(
135   inflation_variables,
136   data = pdata_inflation,
137   model = "within",
138   effect = "individual"
139 )
140
141 model_inflation_re_oneway <- plm(
142   inflation_variables,

```

```

142   data = pdata_inflation,
143   model = "random",
144   effect = "individual"
145 )
146
147 summary(model_inflation_ols)
148 summary(model_inflation_fe_oneway)
149 summary(model_inflation_re_oneway)
150
151 hausman_inflation_oneway <- phptest(model_inflation_fe_oneway, model_inflation_re_oneway)
152 hausman_inflation_oneway
153 ### p-value is 0.8203, so RE can be used here for inflation model (one-way)
154
155 ## Twoway FE/RE Effects
156 model_inflation_fe_twoway <- plm(
157   inflation_variables,
158   data = pdata_inflation,
159   model = "within",
160   effect = "twoways"
161 )
162
163 model_inflation_re_twoway <- plm(
164   inflation_variables,
165   data = pdata_inflation,
166   model = "random",
167   effect = "twoways"
168 )
169
170 summary(model_inflation_fe_twoway)
171 summary(model_inflation_re_twoway)
172
173 hausman_inflation_twoway <- phptest(model_inflation_fe_twoway, model_inflation_re_twoway)
174 hausman_inflation_twoway
175 ### p-value is 0.9675, so RE can be used here for inflation model (two-way)
176
177 #####
178
179 ## <<<<<<<< Regression Diagnostics >>>>>>>>
180
181 ## Checking for Multicollinearity with VIF
182 lm_main <- lm(

```

```

183   fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth,
184   data = data_model_main
185 )
186
187 vif_main <- car::vif(lm_main)
188 vif_main
189 ### all vif values are very close to 1, so almost no multicollinearity
190
191 lm_inflation <- lm(
192   fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth + inflation,
193   data = data_model_with_inflation
194 )
195
196 vif_inflation <- car::vif(lm_inflation)
197 vif_inflation
198 ### all vif values are very close to 1, so almost no multicollinearity
199
200 ## Checking for Heteroscedasticity
201 bptest(model_main_re_twoway)
202 bptest(model_inflation_re_twoway)
203
204 ### There is evidence of heteroscedasticity in the main models because the BP p-value is below
0.05.
205 ### There is no statistically significant evidence of heteroscedasticity in the inflation models
at the 5% level.
206
207 ## Checking for Serial Correlation
208 pbgttest(model_main_re_twoway)
209 pbgttest(model_inflation_re_twoway)
210
211 ### All serial correlation tests here have p-values below 0.05, so there is strong evidence of
serial correlation.
212
213 ## Checking for Cross-Sectional Dependence
214 pcptest(model_main_re_twoway, test = "cd")
215 pcptest(model_inflation_re_twoway, test = "cd")
216
217 ## Robustness Check for Main Model
218 lm_main <- lm(
219   fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth,
220   data = data_model_main

```

```

221 )
222 vif_main <- car::vif(lm_main)
223 vif_main
224 bptest(model_main_re_twoway)
225 pbgttest(model_main_re_twoway)
226 pcptest(model_main_re_twoway, test = "cd")
227
228
229 ## ===== Addressing Serial Correlation with AR(1) =====
230 ## Main Model: Two-way RE + AR(1)
231 model_main_re_twoway_ar1 <- lme(
232   fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth + factor(year),
233   random = ~ 1 | isoalpha3,
234   data = data_model_main,
235   correlation = corCAR1(form = ~ year | isoalpha3),
236   method = "ML"
237 )
238 summary(model_main_re_twoway_ar1)
239
240 ## Inflation Model: Twoway RE + AR(1)
241 model_inflation_re_twoway_ar1 <- lme(
242   fdi_gdp ~ dollarization + exchange_rate + trade_gdp + gdp_growth + inflation +
243     factor(year),
244   random = ~ 1 | isoalpha3,
245   data = data_model_with_inflation,
246   correlation = corCAR1(form = ~ year | isoalpha3),
247   method = "ML"
248 )
249 summary(model_inflation_re_twoway_ar1)
250
251 ## With Clustered SEs
252 coef_test(
253   model_main_re_twoway_ar1,
254   vcov = "CR2",
255   cluster = data_model_main$isoalpha3
256 )
257
258 coef_test(
259   model_inflation_re_twoway_ar1,
260   vcov = "CR2",
261   cluster = data_model_with_inflation$isoalpha3

```



```
262 )  
263  
264 ## ACF plots for normalized residuals  
265 acf(resid(model_main_re_twoway_ar1, type = "normalized"),  
266     main = "ACF: Main Two-way RE + AR(1) Normalized Residuals")  
267 acf(resid(model_inflation_re_twoway_ar1, type = "normalized"),  
268     main = "ACF: Inflation Two-way RE + AR(1) Normalized Residuals")
```